

Smart Compaction: Predicting Compaction Utility from Lakehouse Table Metadata[☆]

Jannic Alexander Cutura
ECB[†] & DSTI[‡]

jannic.cutura@dsti.institute

Subash Prakash
IBM[§]

subash.prakash@ibm.com

Abstract—Open lakehouse table formats accumulate small data files over time, which degrades query performance. Deciding when compaction is worthwhile remains threshold-driven, but which metadata features actually determine compaction utility is not well understood. We present an open simulation framework that generates 2,376 Apache Iceberg tables spanning three orders of magnitude in file size, extracts 17 metadata features from manifest files without reading data, and trains XGBoost to predict the continuous file-reduction ratio ($R^2 = 0.998$, RMSE = 0.013). The binary compaction decision turns out to be trivially separable by a single partition-level threshold (`max_files_per_partition` > 4), requiring no learned model. Cross-schema validation on 96 TPC-H tables confirms generalisation without retraining ($R^2 = 0.976$). A query benchmark reveals that compaction benefits metadata-heavy queries but can slow full-scan aggregations by reducing task parallelism. All code and data are publicly available.

Index Terms—data lakehouse, open table formats, compaction, machine learning, table optimisation

I. INTRODUCTION

The lakehouse architecture [1] unifies data warehousing and data lake workloads through open table formats like Apache Iceberg [2], Delta Lake [3], and Apache Hudi [4]. Each transaction appends immutable columnar files, buying snapshot isolation at the cost of progressive file accumulation.

This yields the *small file problem*: file counts grow without bound, inflating query planning latency, per-request object-store costs, and scan overhead. At scale, catalogs can accumulate millions of undersized files [5]. *Compaction* rewrites them into fewer target-sized files via bin-packing, sort, or z-order strategies, but is itself costly in compute and I/O. The question is therefore not *whether* to compact but *when* the file reduction justifies the cost.

[☆]The views expressed in this paper are solely those of the authors and do not necessarily reflect the views of the European Central Bank, the European System of Central Banks or IBM. All code and data for full replication are available at <https://github.com/JannicCutura/smart-compaction>.

[†]European Central Bank, Sonnemannstraße 20, 60314 Frankfurt am Main, Germany.

[‡]DSTI School of Engineering, 59 Rue de l'Abbé Groult, 75015 Paris, France.

[§]IBM, IBM Campus 1, 71139 Ehningen, Germany.

© 2026 IEEE. Personal use of this material is permitted. Permission from IEEE must be obtained for all other uses, in any current or future media, including reprinting/republishing this material for advertising or promotional purposes, creating new collective works, for resale or redistribution to servers or lists, or reuse of any copyrighted component of this work in other works.

Currently, all open-source compaction scheduling is threshold-driven: Iceberg's `rewrite_data_files` is purely imperative [2], Delta Lake triggers on small-file counts [3], and Hudi uses rule-based triggers [4]. AutoComp [5], the most comprehensive automated system to date, uses heuristics and multi-objective Pareto ranking over 35,000 Iceberg tables at LinkedIn; its authors explicitly identify ML-based prediction as future work. Proprietary systems (Databricks Predictive Optimization [6], Snowflake automatic clustering [7], AWS S3 Tables [8]) are not open or reproducible. No published study has examined *which* table metadata features actually determine compaction utility.

This paper addresses the gap. We generate 2,376 Iceberg tables spanning three orders of magnitude in file size, extract 17 metadata features from manifest files without reading data, and evaluate how well these features predict bin-packing compaction outcomes. We make three contributions:

- 1) A reproducible simulation framework and open dataset of 2,376 Iceberg tables with heterogeneous write patterns, Zipfian skew, and file-size heterogeneity—the last being essential to avoid a *tautology trap* in which uniformly small files make outcomes deterministic.
- 2) An XGBoost regressor that predicts the continuous file-reduction ratio ($R^2 = 0.998$, robust across prevalence levels, 30-seed bootstrap splits, and feature subsets). The binary compaction decision, by contrast, reduces to a deterministic threshold (`max_files_per_partition` > 4) and does not require a learned model.
- 3) Cross-schema validation on 96 TPC-H tables confirming generalisation without retraining ($R^2 = 0.976$), plus a query benchmark revealing that compaction benefits metadata-only queries but can slow full-scan aggregations by reducing task parallelism.

A. Compaction in LSM-Trees

LSM-tree research has established merge-policy cost models [9], [10], quantified the $63\times$ data movement overhead [11], and applied RL [12] and dynamic programming [13] to timing decisions. Lakehouse tables differ fundamentally: flat columnar files on object stores with no level hierarchy, where file proliferation rather than write amplification drives cost. We share the core question — *when is reorganising stored data*

worth the compute cost? — but answer it with metadata-driven prediction rather than policy optimisation.

B. Automated Database Tuning and ML for Systems

Database auto-tuning has progressed from rule-based advisors [14] through OtterTune [15] (Gaussian processes) and CDBTune [16] (deep RL) to NoisePage’s [17] forecast-model-plan framework, which our work instantiates: metadata features forecast compaction utility, and the binary decision is the action plan. In the broader “ML for systems” movement [18], Bao [19] steers query optimisation through learned hint selection and Qd-tree [20] learns workload-aware layouts. Yet no published work applies ML to predict the utility of a storage maintenance operation in the lakehouse. AutoComp [5] is the closest predecessor, relying on heuristic trait computation rather than learned models.

II. METHODOLOGY

A. Problem Formulation

We define the *compaction utility* of a table as the file count reduction ratio achieved by running Iceberg’s `rewrite_data_files` with bin-packing:

$$r_{fc} = 1 - \frac{n_{\text{after}}}{n_{\text{before}}} \quad (1)$$

where n_{before} and n_{after} are the number of data files before and after compaction, respectively. A value of $r_{fc} \approx 1$ indicates high compaction benefit; $r_{fc} \approx 0$ means that compaction produces negligible file reduction and the associated I/O cost is wasted.

We address two complementary tasks:

- 1) **Classification.** Predict whether a table *needs compaction*, defined as whether Iceberg’s bin-packing actually rewrites any data files ($n_{\text{rewritten}} > 0$). This label captures compaction *activity* rather than *utility*; however, the distribution is bimodal: 99.5% of positive labels have $r_{fc} > 0.5$ (only 11 of 2,376 tables lie in $(0, 0.5]$), so activity coincides with meaningful file reduction in practice.
- 2) **Regression.** Predict the continuous reduction ratio $r_{fc} \in [0, 1]$ to enable cost-aware scheduling. This is the paper’s primary prediction task.

B. Simulation Framework

Production tables have unknown ground-truth compaction utility, so we construct a simulation that generates Iceberg tables with controlled characteristics, compacts them, and records the outcome.

1) *Parameter Grid:* Table I lists seven configuration axes spanning 2,376 combinations. The axes cover three orders of magnitude in file size (8 KB to 128 MB),¹ partition cardinalities from 1 (unpartitioned) to 100, write parallelism of

¹The `file_size_target_kb` column in Table I specifies the Spark write-side parameter. Actual on-disk file sizes differ because the heterogeneous batch pattern (Section II-B2) uses an $8\times$ smaller target for even-indexed batches, creating within-table variance.

TABLE I
SIMULATION PARAMETER GRID

Parameter	Description	Values
<code>num_rows</code>	Total rows per table	10K, 100K, 1M
<code>num_columns</code>	Payload columns (excl. part. key)	5, 50
<code>num_partitions</code>	Number of partitions (1 = none)	1, 10, 100
<code>num_writers</code>	Writers per batch	1, 5
<code>num_write_batches</code>	Append rounds	1, 5, 20
<code>file_size_target_kb</code>	Target file size (KB)	8, 128, 512, 2K, 4K, 8K, 16K, 24K, 32K, 65K, 131K
<code>partition_skew</code>	Skew profile	uniform, zipf
<i>Full grid: 2376 configurations</i>		

1 or 5 writers, append frequencies from 1 to 20 batches, and uniform vs. Zipfian ($s=1$) partition skew. At 50 UUID columns (~ 1.5 KB/row in Parquet), files can reach the 128 MB compaction target at scale.

2) *Heterogeneous Write Patterns:* To model production heterogeneity, the first batch receives 50% of rows (initial load) while remaining batches split the rest equally. Even-indexed batches use a file-size target $8\times$ smaller than the configured value, creating within-table variance that breaks the tautology.

3) *Compaction Procedure:* We invoke Iceberg’s `rewrite_data_files` with bin-packing: target 128 MB, minimum 96 MB (75%), maximum 230 MB (180%). Files in [96, 230] MB are not rewritten, matching default production configuration.

C. Feature Engineering

We extract 17 observable features exclusively from Iceberg metadata (manifest files, snapshot history, and partition statistics) without reading any data files. This makes feature extraction fast and non-intrusive in production. The feature vector comprises eight file-level statistics (`file_count`, `total_size_bytes`, average/minimum/maximum/standard-deviation of file size, `total_records`, `avg_records_per_file`), five partition-level statistics (`num_partitions_actual`, average/maximum/minimum/standard-deviation of files per partition), one snapshot count (`num_snapshots`), and three derived ratios: `small_file_ratio` (fraction of files below the 96 MB compaction threshold), `file_size_cv` (coefficient of variation of file sizes), and `files_per_partition_cv` (coefficient of variation of files per partition). These ratios capture the fragmentation and heterogeneity that drive compaction decisions. Note that `small_file_ratio` uses the same 96 MB threshold that Iceberg’s bin-packing filter uses to select files for rewriting. This creates a potential target-leakage concern; we discuss its impact in Section III-I and show via ablation (Section III-E)

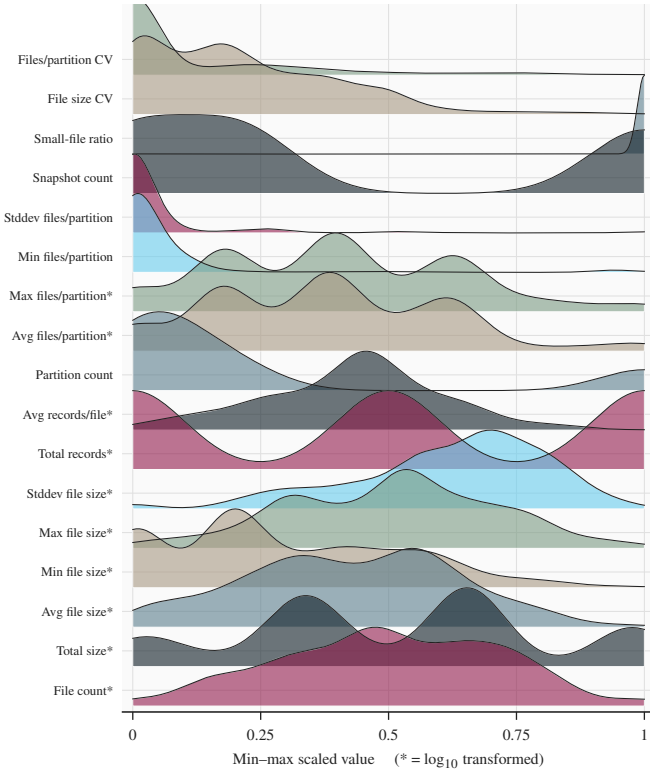


Fig. 1. Ridge plot of the 17 input features, each min-max scaled to $[0, 1]$ for visual comparability. Features marked with an asterisk are $\log_{10}$ -transformed before scaling. Each row shows the kernel density estimate of one feature across all 2,376 tables.

that it is not a top predictor and removing it does not degrade model performance.

Fig. 1 shows the distribution of each feature, split by compaction outcome.

D. Model Selection

We train four models on the 17-feature vector: an XGBoost classifier [21] and logistic-regression baseline for the binary `needs_compaction` task, and an XGBoost regressor and OLS baseline for continuous r_{fc} . All use an 80/20 stratified train/test split with 5-fold cross-validation. Classification is evaluated via accuracy, F1, and ROC-AUC; regression via RMSE and R^2 .

III. EXPERIMENTAL EVALUATION

A. Dataset Overview

The simulation produces 2,376 tables. Of these, 2,121 (89.3%) have at least one rewritten data file and are labelled `needs_compaction = 1`. The file-reduction ratio ranges from 0.000 to 0.999 (mean = 0.818, median = 0.950, $\sigma = 0.294$), exhibiting a heavy left tail of tables that gain little from compaction.

B. Binary Compaction Decision

Before addressing the regression task, we establish that the binary compaction decision does not require a learned

TABLE II
REGRESSION RESULTS ON THE HELD-OUT TEST SET.

Model	RMSE	MAE	R^2
XGBoost	0.013	0.002	0.998
Linear Reg.	0.236	0.159	0.361

model. XGBoost achieves perfect classification ($F1 = 1.000$, $AUC = 1.000$) on the held-out test set ($n = 476$), but so does the trivial threshold `max_files_per_partition` > 4 . A threshold sweep reveals the mechanics: under bin-packing (target 128 MB, min 96 MB), every partition with ≥ 5 files contains at least one eligible for rewriting, so the boundary falls exactly at $k = 4$. This is a deterministic property of the compaction filter, not a learned pattern, and the threshold transfers perfectly to TPC-H (Section III-D).

Feature importance analysis (Fig. 2) confirms extreme concentration: `max_files_per_partition` (gain = 0.676) and `avg_files_per_partition` (gain = 0.188) account for 86.4% of classifier gain and 95.2% of regressor gain. Partition-level file counts dominate both tasks.

C. Regression

The continuous prediction task is where ML adds genuine value. Table II summarises regressor performance. XGBoost achieves $R^2 = 0.998$ (RMSE = 0.013, MAE = 0.002), with a bootstrap 95% CI of $R^2 \in [0.996, 1.000]$ (Section III-G), while OLS captures only modest variance ($R^2 = 0.361$), confirming that a linear model is insufficient for this task. Fig. 3 plots predicted vs. actual reduction ratios.

The OLS shortfall is explained by the nonlinear relationship between files-per-partition and the reduction ratio: the bin-packing target file size creates a saturation curve: once partitions contain enough small files to fill a target file, additional files yield diminishing marginal reduction. XGBoost’s tree splits capture this inflection naturally; a linear model cannot.

The Spearman correlation between `avg_files_per_partition` and r_{fc} is $\rho = 0.965$, confirming that the relationship is monotonic but not linear.

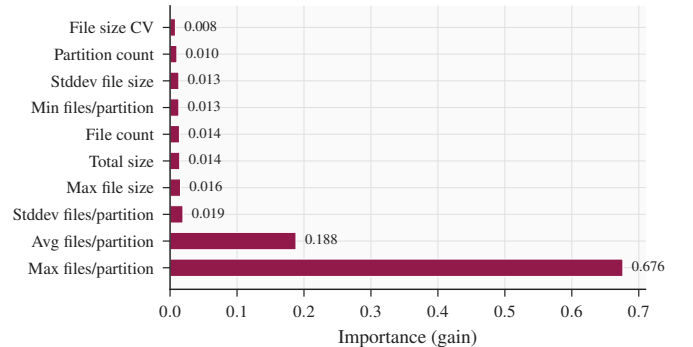


Fig. 2. XGBoost feature importance (gain). Two partition-level features dominate: 86.4% of classifier gain, 95.2% of regressor gain.

TABLE III
PER-BIN REGRESSION ERROR ON THE HELD-OUT TEST SET.

r_{fc} bin	n	RMSE	MAE	Max $ e $
$r_{fc} = 0$	51	0.0039	0.0011	0.026
(0, 0.5]	2	0.0066	0.0064	0.008
(0.5, 0.8]	106	0.0272	0.0074	0.186
(0.8, 0.95]	201	0.0027	0.0011	0.018
(0.95, 1]	235	0.0010	0.0005	0.008

Per-quantile error analysis. Because the r_{fc} distribution is heavily right-skewed (median = 0.950, mean = 0.818), the global RMSE of 0.013 could mask larger errors in sparsely populated bins. Table III stratifies the test set by r_{fc} range. The hardest bin is (0.5, 0.8] (RMSE = 0.027, max $|e|$ = 0.186), where the bin-packing saturation curve is steepest and small feature changes cause large shifts in reduction ratio. Even in this worst case, MAE remains below 0.008, and the remaining bins have RMSE $\leq$ 0.007. The model’s accuracy is therefore consistent across the full target range.

D. Cross-Schema Generalisation

The preceding analysis trains and tests on tables sharing the same synthetic schema (50 UUID columns). To test whether the learned decision boundary transfers across data types and schemas, we evaluate the trained models without retraining on 96 Iceberg tables derived from six TPC-H [22] schemas (lineitem, orders, customer, part, partsupp, supplier), the same benchmark used by Jain et al. [23] for cross-format lakehouse evaluation and by LST-Bench [24] for log-structured table benchmarking. These tables use realistic data types (strings, dates, decimals, and integers) with 7–16 columns per schema instead of the uniform 50-column UUID payload, and are generated at approximately TPC-H scale factor 10 (468.8M total rows, 41,282 files, 15 GB pre-compaction).

To ensure non-trivial compaction outcomes, row counts are scaled so that files span Iceberg’s default 96 MB compaction threshold [2]. The resulting label balance is 79 tables needing compaction (82.3%) vs. 17 that do not.

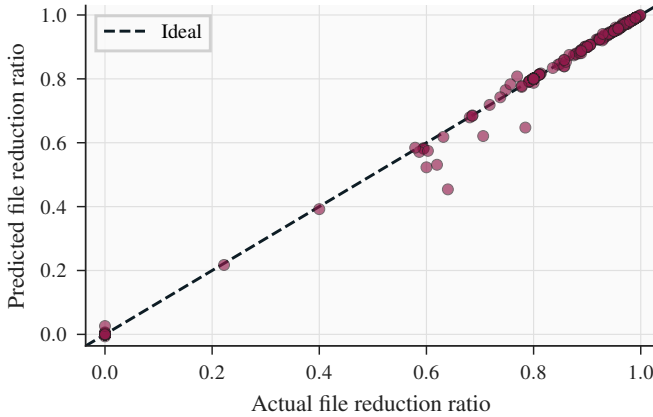


Fig. 3. Predicted vs. actual file reduction ratio on the held-out test set. Points near the dashed line indicate accurate predictions.

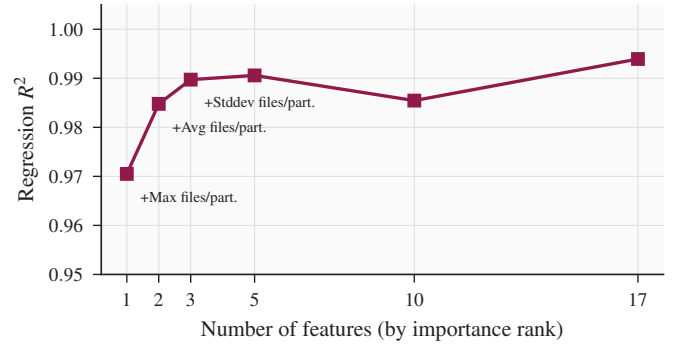


Fig. 4. Regression R^2 vs. the number of top- k features. Two features capture 98.5% of regression variance.

Table VI summarises the results. The regressor achieves $R^2 = 0.976$ (RMSE = 0.058), a moderate degradation from the training distribution ($R^2 = 0.998$) that confirms the model captures transferable structure rather than artefacts of the synthetic schema. The $k=4$ threshold transfers to TPC-H at 97.9% accuracy, with only two false negatives—both borderline cases (3 files at $\sim$ 50MB average, $r_{fc} = 0.333$) where the compactor merged 3 files into 2.

E. Feature Ablation

To determine how many features are necessary and where information concentrates, we conduct two complementary ablation studies.

Progressive top- k . We train XGBoost using only the top- k features ranked by gain importance from the full model (Fig. 4). A single feature (max_files_per_partition) achieves $R^2 = 0.970$. Adding avg_files_per_partition raises R^2 to 0.985; the top-3 features reach 0.990. Beyond five features returns diminish, and ten features slightly degrade regression ($R^2 = 0.985$), consistent with mild overfitting from irrelevant splits.

Leave-one-out. All single-feature removals maintain $R^2 \geq 0.988$, showing no feature is individually indispensable when the remaining 16 are available.

To formally quantify the marginal value of features 3–17, we bootstrap 50 random train/test splits and compute the paired difference $\Delta R^2 = R^2_{k=17} - R^2_{k=2}$ on each. The mean $\Delta R^2 = 0.023$ with 95% CI [0.004, 0.044]; the CI excludes zero, confirming a statistically significant but practically small gain.

These results empirically confirm the feature dominance observed in Section III-B: partition-level file counts carry nearly all the predictive signal. The 15 auxiliary features contribute a statistically significant but small marginal improvement ($\Delta R^2 \approx 0.02$), serving primarily as partial proxies when file count features are noisy or unavailable.

F. Hyperparameter Sensitivity and Model Families

We evaluate robustness to XGBoost configuration by sweeping max_depth $\in \{2, 4, 6, 8\}$ and n_estimators $\in$

TABLE IV
MODEL FAMILY COMPARISON (DEFAULT HYPERPARAMETERS).

Model	Clf Acc.	Clf F1	Reg R^2	Reg RMSE
XGBoost (ours)	1.000	1.000	0.998	0.013
Random Forest	1.000	1.000	0.998	0.013
Decision Tree	0.996	0.998	0.997	0.016
Linear [†]	0.815	0.885	0.361	0.236

[†]Logistic regression for the classification columns, ordinary least squares for the regression columns.

TABLE V
PREVALENCE ROBUSTNESS EVALUATION.

Prev.	n	XGB Acc.	$k=4$ Acc.	Reg R^2
89.3%	2,376	1.000	1.000	0.998
70%	849	1.000	0.989	0.995
50%	510	1.000	0.982	0.998
30%	364	1.000	0.975	0.997

XGBoost and the threshold baseline ($k=4$) evaluated on downsampled datasets; regression uses XGBoost.

{10, 50, 200, 500} (16 configurations). Regression is sensitive to capacity: depth-2 with 10 trees yields $R^2 = 0.852$, but depth-2 with 50 trees already reaches $R^2 = 0.994$. Configurations with depth ≥ 4 and ≥ 50 estimators saturate at $R^2 \geq 0.997$.

To verify that the finding is not XGBoost-specific, we compare four model families (Table IV). Decision trees and random forests achieve near-perfect metrics, confirming that any tree-based model suffices. The linear baselines perform markedly worse on both tasks—logistic regression reaches only F1 = 0.885 on the binary decision, and OLS attains $R^2 = 0.361$ on the regression target—underscoring the non-linear structure of r_{fc} .

G. Statistical Confidence and Prevalence Robustness

Bootstrap confidence intervals. To guard against split-dependent results, we repeat the 80/20 stratified split 30 times with different random seeds and report 95% percentile bootstrap CIs. The regressor achieves $R^2 = 0.999$ [0.996, 1.000] and RMSE = 0.009 [0.004, 0.019].

Prevalence sensitivity. The 89.3% positive prevalence raises legitimate concern that metrics are inflated by class imbalance. We downsample the majority class to create balanced evaluation sets at 70%, 50%, and 30% positive prevalence (Table V). Regression R^2 stays above 0.995 at all levels, confirming that model performance is not an artefact of class imbalance. The $k=4$ threshold degrades slightly at lower prevalence (97.5% at 30%) but remains competitive for the binary decision.

Table VI(b) breaks down performance by TPC-H schema. Four schemas achieve near-perfect regression ($R^2 > 0.997$). Customer and partsupp show lower metrics ($R^2 \approx 0.87$, $n = 16$ each), likely because these schemas produce borderline tables near the compaction threshold.

H. Query Performance Impact

To validate that structural compaction utility translates to downstream benefit, we benchmark query latency before and after compaction on 52 TPC-H tables using Iceberg time-travel (VERSION AS OF) to query both snapshots non-destructively. Each query is run with 1 warmup and 3 timed iterations; we report the median.

Table VII summarises the results by query type. Full-scan COUNT(*) queries benefit consistently from compaction (median $1.19\times$ speedup), with higher gains on tables with more files (lineitem: $1.41\times$ at $\sim 1,200$ pre-compaction files). The Spearman correlation between file-reduction ratio and COUNT(*) speedup is $\rho = 0.50$ ($p < 0.001$), confirming that metadata overhead (file listing, Parquet footer reads) dominates this query type.

Filtered aggregation queries show mixed results. The date-filtered orders_agg achieves $1.00\times$ (neutral), while tpch_q1 on lineitem shows $0.76\times$ (24% slower). Full-scan aggregations without predicates (cust_nation_agg, part_type_agg) are consistently slower after compaction ($0.46\times$ and $0.45\times$ respectively). We attribute this to reduced task parallelism: Spark assigns tasks proportional to file count, so compacting 300 files into 1 reduces parallelism from 300 tasks to 1, offsetting the metadata benefit for queries that must read all data. This finding is consistent with observations in LST-Bench [24] that compaction effects are workload-dependent.

The overall median speedup across all 104 measurements is $0.97\times$ (mean $1.08\times$), indicating that bin-packing compaction is not universally beneficial for query performance (Fig. 5). This reinforces the value of a learned compaction scheduler: rather than compacting all tables with small files, a production system should consider query workload when prioritising compaction, using the regression prediction as a structural cost signal and the query-type mix to estimate downstream benefit.

TABLE VI
CROSS-SCHEMA VALIDATION ON 96 TPC-H TABLES (NO RETRAINING).

(a) Aggregate results					
Task / Model		Acc.	F1	RMSE	R^2
Classification					
XGBoost		0.979	0.987	—	—
Threshold ($k=4$)		0.979	0.987	—	—
Threshold ($sfr>0$)		0.833	0.908	—	—
Majority class		0.823	0.903	—	—
Regression					
XGBoost		—	—	0.058	0.976
(b) Per-schema breakdown					
Schema	n	Clf Acc.	Clf F1	Reg R^2	Reg RMSE
lineitem	16	1.000	1.000	0.997	0.019
orders	16	1.000	1.000	1.000	0.003
part	16	1.000	1.000	1.000	0.002
supplier	16	1.000	1.000	1.000	0.002
customer	16	0.938	0.966	0.874	0.097
partsupp	16	0.938	0.966	0.867	0.101

TABLE VII
QUERY LATENCY SPEEDUP AFTER COMPACTION.

Query	n	Med.	Mean	ρ	p
count_star	52	1.19×	1.31×	0.50	<0.001
orders_agg	12	1.00×	1.63×	0.89	<0.001
tpch_q1	13	0.76×	0.82×	0.77	0.002
cust_nation	15	0.46×	0.56×	0.45	0.095
part_type	12	0.45×	0.47×	0.38	0.226

52 TPC-H tables; median of 3 timed runs.
Speedup > 1× = faster post-compaction.

I. Discussion

1) *Label Definition and Feature Leakage*: The binary label `needs_compaction` captures whether the compactor *rewrote* any files, not whether the resulting reduction was practically beneficial. This distinction matters in principle, but our dataset mitigates it: the r_{fc} distribution is bimodal (mean = 0.818, median = 0.950), with 99.5% of positive-label tables exhibiting $r_{fc} > 0.5$. A table that triggers compaction almost always achieves substantial file reduction. Regardless, the paper’s primary contribution is the regression model, which predicts r_{fc} directly and does not depend on the binary label.

The feature `small_file_ratio` (fraction of files below 96 MB) uses the same threshold as the compaction filter, raising a target-leakage concern. However, its contribution is negligible in both tasks: it carries only 0.009 gain in the regressor — against 0.795 for `max_files_per_partition` — and does not enter the classifier’s top ten at all. The leave-one-out ablation (Section III-E) confirms this: removing it does not degrade R^2 below 0.988. The leakage is therefore present but inconsequential.

2) *Operational Cost-Benefit Threshold*: A production scheduler needs more than a predicted r_{fc} —it needs a threshold below which compaction is not worth the I/O cost. Our dataset provides empirical guidance: of the 2,121 tables that trigger compaction, only 11 (0.5%) have $r_{fc} \leq 0.5$, and tables with $r_{fc} \leq 0.1$ eliminate zero files on average. A natural scheduling rule is therefore: compact only when the predicted $r_{fc} > 0.3$, which would avoid the four lowest-

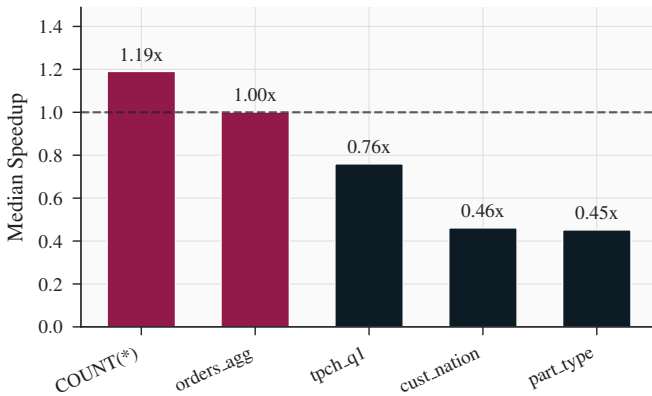


Fig. 5. Median query speedup by query type after compaction.

utility compactations while retaining 99.8% of beneficial ones. The precise threshold is deployment-specific (it depends on storage cost, I/O bandwidth, and query patterns), but the regressor provides the continuous signal needed for cost-aware prioritisation.

3) *Tautology Trap*: An earlier parameter grid produced only small files (max ~0.4 MB vs. the 96 MB threshold), so *every* table compacted completely: $r_{fc} = 1 - 1/N$. XGBoost achieved $R^2 = 0.99$ by recovering this closed-form identity (Spearman $\rho = 0.996$ between `avg_files_per_partition` and the target). File-size heterogeneity breaks this relationship because the bin-packing filter skips files in [96, 230] MB, forcing the model to learn non-trivial interactions.

This finding has practical implications: any compaction benchmark that uses only small synthetic files will produce misleadingly perfect ML metrics. Evaluation must include files spanning the compaction target range to create non-trivial prediction targets, a concern echoed in LST-Bench’s design of mixed-size workloads [24].

4) *Feature Dominance*: The concentration of importance in files-per-partition features is not a modelling artefact but reflects the mechanics of bin-packing compaction (1). The compaction procedure [2] rewrites files smaller than the minimum threshold (96 MB) and bins them into target-sized outputs (128 MB). The number of eligible small files per partition therefore determines both the binary outcome (are there files to rewrite?) and the reduction magnitude.

This dominance has a direct operational consequence: AutoComp’s [5] compaction traits already include partition-level file counts as one of several heuristic inputs. Our ablation study (Section III-E) provides quantitative evidence that these traits alone are sufficient for the binary scheduling decision, while Section III-F shows the result holds across model families (decision trees, random forests). More sophisticated ML models are only justified when predicting the continuous reduction ratio for cost-aware scheduling.

5) *Threats to Validity*: Synthetic UUID payloads compress differently from real columnar data [25]. Partition structure is generated, not domain-derived; query patterns are not modelled. Spark-local execution may differ from distributed clusters with HDFS or S3, a gap well-documented in database tuning research [26]. The 89.3% positive class prevalence risks inflating accuracy metrics. We address this with three mitigations: (i) a majority-class baseline (Table VI), (ii) prevalence downsampling experiments showing stable regression at 50% balance (Section III-G), and (iii) bootstrap CIs over 30 random splits (Section III-G). The TPC-H cross-schema experiment (Section III-D) partially mitigates the synthetic-data concern by validating on six relational schemas with realistic data types, but production tables with schema evolution, delete files, and mixed CoW/MoR strategy [27] remain untested.

Three additional threats deserve explicit acknowledgement. *Delete files and Merge-on-Read (MoR)*. Our simulation uses Copy-on-Write exclusively. MoR tables accumulate positional and equality delete files that compaction must merge into

data files — a fundamentally different compaction dynamic not captured by our feature set. *Sort and z-order compaction.* We evaluate bin-packing only. Sort-based compaction rewrites files for query locality; its utility depends on query workload patterns, which a metadata-only approach cannot capture. The title “Predicting Compaction Utility” therefore applies to bin-packing utility specifically. *Temporal evolution.* Our model operates on static snapshots. Production tables evolve continuously; a table that does not need compaction now may need it after the next batch of writes. Temporal prediction (e.g., forecasting compaction need k commits ahead) is future work.

6) *Compaction Target Sensitivity:* All experiments use the same compaction configuration (target 128 MB, minimum 96 MB, maximum 230 MB). The $k=4$ threshold is implicitly tied to this target: a larger target size (e.g., 512 MB, Iceberg’s production default) would produce fewer output files per partition, likely shifting the optimal k downward. The learned XGBoost model may generalise better than a fixed threshold because it incorporates multiple features (file sizes, counts, ratios), but this is untested. Evaluating sensitivity to compaction target size is a priority for future work.

7) *Compaction Cost:* Although our framework collects wall-clock compaction duration, this study focuses on predicting *structural* compaction outcomes (file count reduction) rather than *economic* cost. A production cost model would additionally require cloud I/O pricing, compute cost, and downstream query speedup estimates. We observe a moderate rank correlation between compaction duration and the number of files rewritten (Spearman $\rho = 0.76$ on synthetic data, $\rho = 0.72$ on TPC-H), suggesting that the regression prediction can serve as a rough proxy for cost ranking, but a full cost-benefit analysis following Cosine [28] remains future work.

IV. CONCLUSION

We studied the predictability of bin-packing compaction utility in Apache Iceberg using 17 metadata features across 2,376 synthetic tables and validated on 96 TPC-H tables spanning six relational schemas. The binary compaction decision turns out to be a deterministic property of the bin-packing filter, solvable by a single threshold (`max_files_per_partition` > 4) without any learned model. The genuine ML contribution lies in predicting the continuous file-reduction ratio: XGBoost achieves $R^2 = 0.998$ (CI [0.996, 1.000]), generalises to TPC-H without retraining ($R^2 = 0.976$), and is robust across prevalence levels, feature subsets, and model families.

Query benchmarking on 52 TPC-H tables reveals that compaction’s downstream impact is workload-dependent: metadata-heavy queries benefit ($1.19\times$ median `COUNT(*)` speedup), while full-scan aggregations suffer from reduced task parallelism ($0.45\text{--}0.76\times$). This motivates workload-aware compaction scheduling: the regression prediction provides the structural cost signal, but a production system must also consider query patterns when prioritising compaction.

Future work includes: (i) extending the framework to Delta Lake and Hudi; (ii) incorporating query-workload features

and cloud-cost metrics following Cosine [28]; (iii) multi-table scheduling via contextual bandits [19]; (iv) sort and z-order compaction strategies; and (v) validation on production Iceberg catalogs.

ACKNOWLEDGMENT

The authors thank the open-source Apache Iceberg and Apache Spark communities for providing the software infrastructure used in this research. We thank DSTI School of Engineering for research support.

REFERENCES

- [1] M. Armbrust, A. Ghodsi, R. Xin, and M. Zaharia, “Lakehouse: A new generation of open platforms that unify data warehousing and advanced analytics,” in *Proceedings of the 11th Conference on Innovative Data Systems Research (CIDR)*, 2021. [Online]. Available: http://cidrdb.org/cidr2021/papers/cidr2021_paper17.pdf
- [2] Apache Software Foundation, “Apache Iceberg table format specification,” <https://iceberg.apache.org/spec/>, 2024, open table format specification, originally developed at Netflix.
- [3] M. Armbrust, T. Das, L. Sun, B. Yavuz, S. Zhu, M. Murthy, J. Torres, H. van Hovell, A. Ionescu, A. Łuszczak, M. Świtakowski, M. Szafranski, X. Li, T. Ueshin, M. Mokhtar, P. Boncz, A. Ghodsi, S. Paranjpye, P. Senster, R. Xin, and M. Zaharia, “Delta lake: High-performance ACID table storage over cloud object stores,” *Proceedings of the VLDB Endowment*, vol. 13, no. 12, pp. 3411–3424, 2020.
- [4] Apache Software Foundation, “Apache Hudi compaction,” <https://hudi.apache.org/docs/compaction/>, 2024, documentation of Hudi compaction triggers and strategies.
- [5] A. Gruenheid, J. Camacho-Rodríguez, C. Curino, R. Ramakrishnan, S. Pak, S. Sakdeo, L. Gandhi, S. K. Singhal, P. Nilangekar, and D. J. Abadi, “AutoComp: Automated data compaction for log-structured tables in data lakes,” in *Companion of the 2025 International Conference on Management of Data (SIGMOD/PODS ’25)*, 2025, pp. 404–417.
- [6] Databricks, “Predictive optimization for Unity Catalog managed tables,” <https://docs.databricks.com/aws/en/optimizations/predictive-optimization>, 2024, ai/ML-driven automatic table maintenance.
- [7] Snowflake Inc., “Automatic clustering,” <https://docs.snowflake.com/en/user-guide/tables-auto-reclustering>, 2024.
- [8] Amazon Web Services, “How Amazon S3 Tables use compaction to improve query performance,” <https://aws.amazon.com/blogs/storage/how-amazon-s3-tables-use-compaction-to-improve-query-performance-by-up-to-3-times/>, 2025.
- [9] N. Dayan, M. Athanassoulis, and S. Idreos, “Monkey: Optimal navigable key-value store,” in *Proceedings of the 2017 ACM International Conference on Management of Data (SIGMOD)*, 2017, pp. 79–94.
- [10] N. Dayan and S. Idreos, “Dostoevsky: Better space-time trade-offs for LSM-tree based key-value stores via adaptive removal of superfluous merging,” in *Proceedings of the 2018 ACM International Conference on Management of Data (SIGMOD)*, 2018, pp. 505–520.
- [11] S. Sarkar, D. Staratzis, Z. Zhu, and M. Athanassoulis, “Constructing and analyzing the LSM compaction design space,” *Proceedings of the VLDB Endowment*, vol. 14, no. 11, pp. 2216–2229, 2021.
- [12] D. Mo, F. Chen, S. Luo, and C. Shan, “Learning to optimize LSM-trees: Towards a reinforcement learning based key-value store for dynamic workloads,” *Proceedings of the ACM on Management of Data*, vol. 1, no. 3, 2023.
- [13] C. Luo, S. Chatterjee, R. Ketsetsidis, K. Nagaraj, M. J. Amiri, and S. Idreos, “Rethinking the compaction policies in LSM-trees,” *Proceedings of the ACM on Management of Data*, 2025, sIGMOD 2025.
- [14] S. Chaudhuri and V. Narasayya, “Self-tuning database systems: A decade of progress,” in *Proceedings of the 33rd International Conference on Very Large Data Bases (VLDB)*, 2007, pp. 3–14.
- [15] D. Van Aken, A. Pavlo, G. J. Gordon, and B. Zhang, “Automatic database management system tuning through large-scale machine learning,” in *Proceedings of the 2017 ACM International Conference on Management of Data (SIGMOD)*, 2017, pp. 1009–1024.

- [16] J. Zhang, Y. Liu, K. Zhou, G. Li, Z. Xiao, B. Cheng, J. Xing, Y. Wang, T. Cheng, L. Liu, M. Ran, and Z. Li, "An end-to-end automatic cloud database tuning system using deep reinforcement learning," in *Proceedings of the 2019 ACM International Conference on Management of Data (SIGMOD)*, 2019, pp. 415–432.
- [17] A. Pavlo, M. Butrovich, L. Ma, W. S. Lim, P. Menon, D. Van Aken, and W. Zhang, "Make your database system dream of electric sheep: Towards self-driving operation," *Proceedings of the VLDB Endowment*, vol. 14, no. 12, pp. 3211–3221, 2021.
- [18] X. Zhou, C. Chai, G. Li, and J. Sun, "Database meets artificial intelligence: A survey," *IEEE Transactions on Knowledge and Data Engineering*, vol. 34, no. 3, pp. 1096–1116, 2022.
- [19] R. Marcus, P. Negi, H. Mao, N. Tatbul, M. Alizadeh, and T. Kraska, "Bao: Making learned query optimization practical," in *Proceedings of the 2021 ACM International Conference on Management of Data (SIGMOD)*, 2021, pp. 1275–1288, best Paper Award.
- [20] Z. Yang, B. Chandramouli, C. Wang, J. Gehrke, Y. Li, U. F. Minhas, P.-Å. Larson, D. Kossmann, and R. Acharya, "Qd-tree: Learning data layouts for big data analytics," in *Proceedings of the 2020 ACM SIGMOD International Conference on Management of Data*, 2020, pp. 193–208.
- [21] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [22] Transaction Processing Performance Council, "TPC-H benchmark specification, revision 3.0.1," TPC, Tech. Rep., 2024. [Online]. Available: <https://www.tpc.org/tpch/>
- [23] P. Jain, P. Kraft, C. Power, T. Das, I. Stoica, and M. Zaharia, "Analyzing and comparing lakehouse storage systems," in *Proceedings of the 13th Conference on Innovative Data Systems Research (CIDR)*, 2023. [Online]. Available: <https://www.cidrdb.org/cidr2023/papers/p92-jain.pdf>
- [24] J. Camacho-Rodríguez *et al.*, "LST-Bench: Benchmarking log-structured tables in the cloud," *Proceedings of the ACM on Management of Data*, vol. 2, no. 1, 2024.
- [25] M. Stonebraker, D. J. Abadi, A. Batkin, X. Chen, M. Cherniack, M. Ferreira, E. Lau, A. Lin, S. Madden, E. O’Neil, P. O’Neil, A. Rasin, N. Tran, and S. Zdonik, "C-Store: A column-oriented DBMS," in *Proceedings of the 31st International Conference on Very Large Data Bases (VLDB)*, 2005, pp. 553–564.
- [26] D. Van Aken, D. Yang, S. Brillard, A. Fiorino, B. Zhang, C. Billian, and A. Pavlo, "An inquiry into machine learning-based automatic configuration tuning services on real-world database management systems," *Proceedings of the VLDB Endowment*, vol. 14, no. 7, pp. 1241–1253, 2021.
- [27] A. Okolnychyi, C. Sun, A. Tanimura, R. Spitzer, R. Blue, S. Ho, P. Gu, A. Lakkundi, and G. Tsai, "Petabyte-scale row-level operations in data lakehouses," *Proceedings of the VLDB Endowment*, vol. 17, no. 12, pp. 4159–4172, 2024.
- [28] S. Chatterjee, M. Jagadeesan, W. Qin, and S. Idreos, "Cosine: A cloud-cost optimized self-designing key-value storage engine," *Proceedings of the VLDB Endowment*, vol. 15, no. 1, pp. 112–126, 2022.